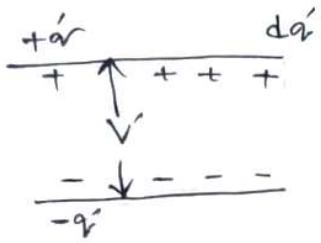
☞ Energy stored in a charged capacitor.

Or, Proove that, $u = \frac{1}{2} \epsilon_0 E^2$, where symbols are their usual meaning.

Answer:



Let q' be the charge and v' the potential difference established between the plates of the capacitor at any instant during the process of charging. If an additional charge dq' is given to the plates, the work done by the battery is given by $dw = v' dq'$

$\because V' = \frac{q'}{C}$

$= \frac{q'}{C} dq'$

Total work done to charge a capacitor to a charge q is

$$w = \int_0^q dw = \int_0^q \frac{q'}{C} dq'$$

$$= \frac{1}{C} \int_0^q q' \, dq'$$

$$\therefore w = \frac{1}{2} \frac{q^2}{C}$$

This work done is stored as electrostatic potential energy in the capacitor. is

$$U = \frac{1}{2}\frac{q^2}{C}$$

$$= \frac{1}{2}\frac{C^2V^2}{C}$$

$$\therefore U = \frac{1}{2}CV^2$$

Consider a parallel plate capacitor of area A and plate separation d, The energy of capacitor,

$$U = \frac{1}{2}CV^2 = \frac{1}{2}\left(\frac{\epsilon_0 A}{d}\right)\cdot V^2$$

Volume of the space between the plate = Ad.
Energy density u is the potential energy per unit volume,

$$u = \frac{U}{Ad}$$

$$= \frac{\frac{1}{2}\left(\frac{\epsilon_0 A}{d}\right)V^2}{Ad}$$

$$= \frac{\epsilon_0 A V^2}{2d} \times \frac{1}{Ad}$$

$$\therefore u = \frac{1}{2}\epsilon_0\left(\frac{V}{d}\right)^2$$

$$\because E = \frac{V}{d}$$

or, $$u = \frac{1}{2}\epsilon_0 E^2$$

Where, u = Energy density.